

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A.  $2.48 \times 10^{-8}$  cm
- B.  $5.92 \times 10^7$  cm/sec
- C.  $8.66 \times 10^{-6}$  cm
- D.  $3.14 \times 10^{-5}$  cm
- E.  $9.81 \times 10^{-4}$  cm
- F.  $5.27 \times 10^{-9}$  m
- G.  $1.23 \times 10^{-7}$  cm
- H.  $4.14 \times 10^{-10}$  m
- I.  $6.625 \times 10^{-27}$  erg sec
- J.  $0.512 \times 10^6$ eV

Reasoning + Answer:

<think>

- First , recall the relation between energy and wavelength in the de Broglie equation :  $E = hv = (hc/\lambda)$  , where h is Planck's constant , c is the speed of light , and  $\lambda$  is the wavelength .
- Convert 1 eV to Joules . 1 eV is approximately  $1.602 \times 10^{-19}$  J .
- Use the relation  $E = hc/\lambda$  to solve for  $\lambda$  .  $\lambda = hc/E$  .
- $h = 6.626 \times 10^{-34}$  Js ,  $c = 3 \times 10^8$  m/s .
- Substitute and rearrange the formula to  $\lambda = (6.626 \times 10^{-34} \text{ Js} * 3 \times 10^8 \text{ m/s}) / (1.602 \times 10^{-19} \text{ J})$ .
- Performing the calculation gives the result in meters , which can then be converted to centimeters by multiplying by 100 .

</think>

<answer>  $2.48 \times 10^{-8}$  cm </answer>